

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.6: Plant Growth and Development — Teacher Reference (Pre-filled)

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand	Sub-Strand 1.6: Plant Growth and Development
Driving Question	DRIVING QUESTION: Why do seeds from the same garden grow and develop so differently, and what controls how a plant grows from a tiny seed into a tall tree or a flowering herb?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Seed Structure and Germination Conditions	Students examined soaked maize and bean seeds (common in Kenyan shambas) and dissected them to identify the testa, cotyledons, embryo (plumule and radicle), and endosperm. They also observed two sets of seeds — one placed in moist cotton wool in a warm spot near a Nairobi classroom window and one kept dry and cold — over 48 hours, noting which set began to germinate.	A seed contains a pre-formed embryo and a food store (endosperm or cotyledons). Germination requires three external conditions: adequate water (imbibition triggers enzyme activation), suitable warmth (optimum ~25–30 °C for most Kenyan crop seeds), and oxygen (for aerobic respiration to release energy). Light is NOT a requirement for germination itself. Seeds kept dry or cold do not germinate because metabolic processes cannot be activated without moisture and warmth.	The differently growing seeds in the driving-question garden may simply have received different amounts of water or been exposed to different soil temperatures — factors that control whether and how quickly germination begins. Teacher note: Reinforce that dormancy is a survival adaptation; maize farmers in the Rift Valley store seed grain in dry conditions deliberately to prevent premature germination.

Lesson 2 Stages of Germination: Epigeal vs. Hypogeal	Students observed time-lapse photographs and live specimens (day 1–7) of germinating bean seeds (epigeal) and maize seeds (hypogeal). They sketched and labelled each stage: water uptake → radicle emergence → shoot emergence → first true leaves. They compared how the cotyledons behaved differently in the two types — bean cotyledons rose above soil; maize scutellum stayed below.	Germination follows a sequence: (1) imbibition of water, (2) activation of amylase and other enzymes that digest stored starch/protein, (3) radicle emerges first (positive geotropism) to anchor the plant and absorb water, (4) plumule emerges and grows upward (negative geotropism). In epigeal germination (beans, pawpaw) the hypocotyl elongates and pulls cotyledons above ground. In hypogeal germination (maize, millet — staples across western Kenya) the cotyledon/scutellum remains below ground and the epicotyl pushes the shoot up. Seedling establishment ends when the plant becomes photosynthetically self-sufficient.	The type of germination partly explains early growth differences: an epigeal seedling (e.g., bean) is vulnerable to soil crust because it must pull cotyledons through; a hypogeal seedling (e.g., maize) is more protected. This helps answer why bean seedlings in a heavy-clay Kisumu garden may struggle while maize seedlings in the same garden push through more easily. Teacher note: Ask students to connect radicle-first emergence to the function of roots before shoots — a classic engineering-practice argument from evidence opportunity.
Lesson 3 Regions of Growth in Roots and Shoots	Students used hand lenses to examine the tips of germinating bean roots and shoots. They observed or were shown prepared slides (or clear diagrams) of the root tip showing: root cap, meristematic (cell division) zone, elongation zone, and maturation zone with root hairs. A simple experiment marked equal 2 mm intervals along a young root with ink; after 24 hours students measured which intervals had expanded most, identifying the elongation zone.	Plant growth is localised, not diffuse. The apical meristem at root and shoot tips contains rapidly dividing undifferentiated cells. Cells produced move back into the elongation zone where they take up water by osmosis and expand — this is what drives root penetration into soil and shoot extension upward. In the maturation zone, cells differentiate into specialised tissues (xylem, phloem, root hair cells). Root hairs massively increase surface area for water and mineral absorption. This is primary growth (increase in length).	Because growth occurs only at specific zones, damage to a root tip (e.g., by a soil pest common in Kenyan farms like the root-knot nematode) can completely halt root elongation even though the rest of the root is intact. This explains uneven plant heights in the driving-question garden — physical or biological damage to the growing tip produces stunted plants beside healthy neighbours. Teacher note: The ink-marking experiment is a powerful NGSS Planning and Carrying Out Investigations practice; ensure students record quantitative data and calculate which zone elongated by what percentage.
Lesson 4 Plant Hormones: Auxins and Phototropism	Students set up or observed the classic oat/bean seedling experiment: one set of seedlings grown with light from one side (unilateral light), one set with tips covered, one set with tips removed. They observed that only intact-tipped seedlings with unilateral light bent toward the light source (positive phototropism). Students also observed photographs of sukuma wiki (kale) grown under a shade net leaning toward the gap — a Kenyan farm context.	Auxin (Indole-3-acetic acid, IAA) is produced at the shoot apex and moves downward. When light strikes from one side, auxin migrates to the shaded side, causing greater cell elongation there, which bends the shoot toward the light. Key points: (1) Auxin promotes elongation in shoots at low concentrations but inhibits elongation at high concentrations. (2) In roots the effect is reversed — even low auxin concentrations inhibit elongation, which is why roots grow away from light (negative phototropism). (3) Removing the tip abolishes the response, proving the tip is the source of the signal.	Plants in a garden shaded by neighbouring maize stalks or a fence will bend toward available light — explaining why some seedlings in the driving-question garden grow at angles while others grow straight. This is an adaptive, hormonally controlled response, not random. Teacher note: Emphasise that auxin is a chemical signal (not nervous-system-based) — a key conceptual distinction for Grade 10 students. Connect to agricultural practice: Kenyan horticulturalists in Thika use synthetic auxins (rooting powders) when propagating rose cuttings.

Lesson 5 Gibberellins, Cytokinins, and Overall Plant Growth Regulation	Students observed (via photographs or demonstration) dwarf maize plants alongside normal maize plants of the same age grown in Kenyan experimental plots. They also observed photographs of seedless grapes (treated with gibberellins) vs. normal grapes, and petri-dish experiments showing cytokinins delaying leaf yellowing (senescence) in cut sukuma wiki leaves stored with and without cytokinin solution.	Gibberellins promote stem elongation (internode extension), seed germination (by activating amylase), and fruit development. Dwarf varieties of maize and wheat lack functional gibberellin pathways — a fact exploited in Kenya's high-yielding dwarf maize breeding programmes. Cytokinins promote cell division, delay senescence, and work antagonistically with auxin to determine shoot-to-root growth balance. Abscissic acid (ABA) promotes dormancy and stomatal closure under drought — highly relevant to Kenya's semi-arid regions. Ethylene promotes fruit ripening and leaf abscission. Together, the ratio and interaction of these hormones — not just any single one — determines the plant's growth pattern.	The driving-question garden seeds grow differently partly because they may be different varieties with different hormonal sensitivities (e.g., a tall local maize variety vs. a dwarf hybrid). Environmental stresses like drought trigger ABA production, slowing growth — explaining why plants near a dry patch of the garden stay small. Teacher note: This lesson is rich for Developing and Using Models (NGSS SEP); students should update their class model to show hormone arrows between plant parts.
Lesson 6 Environmental Factors Affecting Plant Growth	Students analysed data tables and graphs from a school garden investigation (or KARI/KALRO published data) showing maize plant height at harvest under four conditions: full sunlight + adequate water, full sunlight + water stress, shade + adequate water, shade + water stress. They identified trends and anomalies in the data sets and discussed what variables were controlled.	Plant growth is regulated by both internal (hormonal/genetic) and external (environmental) factors. Key environmental factors: (1) Light intensity and photoperiod — affects rate of photosynthesis and, via phytochrome, triggers flowering. (2) Water availability — water is needed for cell turgor (elongation), photosynthesis, and mineral transport; drought stunts growth and triggers ABA-mediated dormancy. (3) Mineral nutrients — nitrogen is essential for protein/chlorophyll synthesis; phosphorus for root development and ATP; potassium for enzyme function and stomatal control. (4) Temperature — affects enzyme reaction rates; frost damages membranes. (5) Soil pH — affects mineral availability. Kenyan Highland soils (Kericho, Meru) are naturally fertile; coastal soils may be sandy and nutrient-poor.	This lesson directly and comprehensively answers the driving question: seeds from the same garden grow differently because they experience micro-variations in all these factors — a patch of shade, a slightly drier or wetter spot, a pocket of compacted or nutrient-depleted soil. The interaction of these factors with internal hormonal signals produces the observed variation. Teacher note: Guide students to distinguish correlation from causation when interpreting the data — a critical NGSS Analysing and Interpreting Data practice. Refer back to earlier lessons on auxin and gibberellins to show the chain: environment → hormone signal → cellular response → observable growth difference.
Lesson 7 Tropisms and Nastic Movements	Students observed (live or via video) the following: (1) A bean seedling root growing around a stone in a pot (thigmotropism / obstacle response). (2) A climbing passion fruit vine tendril wrapping around a support stake (thigmotropism). (3) Photographs of the sensitive plant <i>Mimosa pudica</i> (touch-me-not, found in Kenyan roadsides) leaves	A tropism is a directional growth response to a directional stimulus; the direction of the response depends on the direction of the stimulus (e.g., phototropism, geotropism, hydrotropism, thigmotropism). A nastic movement is a non-directional response to a non-directional stimulus — the direction of movement is determined by plant anatomy,	A climbing passion fruit vine that wraps its tendrils around a wooden post (common in Kenyan kitchen gardens) is using thigmotropism — a hormonally mediated differential elongation response — to access vertical space and light. This is the same mechanism as phototropism studied in Lesson 4, applied to a touch stimulus.

	folding on touch (thigmonasty). (4) A time-lapse of a sunflower head tracking the sun (heliotropism). They classified each movement as a tropism or a nastic movement and identified the stimulus.	not stimulus direction (e.g., thigmonasty in Mimosa, nyctinasty — opening/closing of flowers with day/night). Tropisms involve differential cell elongation (hormones) and are permanent/slow; many nastic movements involve reversible changes in turgor pressure and are rapid. Both are adaptive: tropisms direct organ growth toward resources; nastic movements provide protection or pollination advantage.	The diversity of movement types helps explain why plants with identical genetics can look very different depending on the physical structure of their environment. Teacher note: Ensure students do not confuse nastic movements with tropisms — a common misconception. Use the directionality criterion as the distinguishing test.
Lesson 8 Secondary Growth, Life Cycles, and Flowering	Students examined a cross-section of a tree trunk (a sawn timber offcut, readily available in Kenyan hardware/timber yards) and counted annual rings. They compared the trunk section with a herbaceous stem (e.g., bean or sunflower stem cross-section diagram) to identify the presence of a vascular cambium and cork cambium in woody stems but not herbaceous ones. Students also classified common Kenyan plants (mukau tree, neem, bougainvillea, bean, maize, Napier grass) as annuals, biennials, or perennials, and as monocots or dicots.	Primary growth (from apical meristems) increases length. Secondary growth (from lateral meristems — vascular cambium and cork cambium) increases girth and is found in dicot woody plants. The vascular cambium produces secondary xylem (wood) inward and secondary phloem outward; annual rings reflect seasonal variation in xylem cell size (large in wet season, small in dry season) — particularly visible in trees from Kenya's seasonally dry regions like Tsavo. Herbaceous plants (maize, bean, sukuma wiki) do not undergo secondary growth. Flowering (reproduction) is triggered by photoperiod detected by phytochrome: short-day plants flower when nights exceed a critical length; long-day plants flower when nights are shorter. Life cycle length (annual vs. perennial) is genetically determined but modulated by environment.	The final answer to the driving question integrates all lessons: a tiny maize seed and a mukau tree seed from the same garden will grow so differently because (1) their genetic programmes specify entirely different life cycles and body plans, (2) their hormonal systems respond differently to the same environmental cues, (3) one undergoes only primary growth while the other adds secondary growth year after year, and (4) micro-environmental differences in water, light, temperature, and nutrients amplify genetic differences into dramatically different phenotypes. Teacher note: Use this lesson as a Model Building culmination — students should produce a final annotated diagram connecting seed structure (Lesson 1) → germination (Lesson 2) → meristem growth (Lesson 3) → hormonal control (Lessons 4–5) → environmental factors (Lesson 6) → directional movements (Lesson 7) → secondary growth and life cycle (Lesson 8) to fully answer the driving question.